ASSIGNMENT No:-02

Generate AND/OR function using McCulloch-Pitts neural net by a python program.

In [2]:
```
def mcp(inputs,weights,threshold):
    total=0
    for i in range(len(inputs)):
        total+=inputs[i]*weights[i]

    if total>=threshold:
        return 1
    else:
        return 0

inputs=[[0,0],[0,1],[1,0],[1,1]]

print("AND  Gate")
print("X1 X2 Outputs")
for inp in inputs:
    outputs=mcp(inp,[1,1],2)
    print(inp[0],inp[1],outputs)

print("OR Gate")
print("X1 X2 Outputs")
for inp in inputs:
    outputs=mcp(inp,[1,1],1)
    print(inp[0],inp[1],outputs)
```

```
AND  Gate
X1 X2 Outputs
0 0 0
0 1 0
1 0 0
1 1 1
OR Gate
X1 X2 Outputs
0 0 0
0 1 1
1 0 1
1 1 1
```

In [ ]: